



Rafael Menezes

Postdoctoral Researcher @ ICTP-SAIFR

- +55 71 99669 7729
- r.menezes@ictp-saifr.org
- dos Santos, R. | Menezes, R.
- ICTP-SAIFR / IFT, Unesp, Brazil
- CASUS, HZDR, Germany

Summary

I use deterministic and stochastic models to upscale complex animal behavior and understand its consequences at larger scales.

Links

- www.rafaelmenezes.com
- ORCID
- Research Gate
- @r-menezes (Github)
- Google Scholar

Skills

- Movement Ecology
- Community Ecology
- Complex Networks Theory
- Deterministic/Stochastic Modeling
- Statistical Modeling
- Individual Based Modeling
- Spatial Pattern Analysis
- Monte Carlo Simulation
- Python, R, C++

Employment and Positions

- 2025 – present **Postdoctoral Researcher** ICTP – SAIFR / IFT Unesp
FAPESP Postdoctoral Fellow at ICTP-SAIFR
- 2023 – 2024 **Visiting Researcher** CASUS – HZDR, Germany
Sup: Prof. Dr. Justin Calabrese

Education

- 2021 – 2025 **Ph.D. in Ecology** São Paulo University (USP), Brazil
Advisor: Prof. Dr. Ricardo Martínez-García
Co-Advisor: Prof. Dr. Paulo Inácio Prado
- 2023 – 2025 **MBA in Data Science and Analytics** USP-Esalq, Brazil
- 2017 – 2020 **M.Sc. in Physics** Federal University of Bahia (UFBA), Brazil
Advisor: Profa. Dra. Suani Pinho
Co-Advisors: Profa. Dra. Flora Bacelar, Prof. Dr. Pedro Meirelles
- 2012 – 2017 **B.Sc. and Licenciante in Physics** UFBA, Brazil
- 2014 – 2015 **Exchange Student / Visiting Junior Researcher** Radboud University, NL

Publications

- 2026 **Unveiling the size scaling of extinction and extremal abundance in marine phytoplankton**
Segura, A. M., Cammarota, D., Monteiro, N. Z., Menezes, R., Kruk, C., Perera, G.
Frontiers in Ecology and Evolution, 14, 1832820
- 2025 **The range-resident logistic model: a new framework to formalize the population-dynamics consequences of range residency**
Menezes, R., Calabrese, J., Fagin, W., Prado, P. I., Martinez-Garcia, R.
Ecology Letters, 28(12), e70269
- 2025 **Rare Phyla, Such as CPR and DPANN, Shape Ecosystem-Level Microbial Community Structure Dissimilarities**
Ferreira, C.M., ..., Menezes, R., ..., Meirelles, P.M.
Microbial Ecology, 88(1), 135
- 2025 **Spatial moment dynamics and biomass density equations provide complementary, yet limited, descriptions of pattern formation in individual-based simulations**
Surendran*, A., Pinto-Ramos*, D., Menezes*, R., Martinez-Garcia, R.
Physica D: Nonlinear Phenomena, 477, 134703
- 2023 **Lotka-Volterra model with Allee effect: Equilibria, coexistence and size scaling of maximum and minimum abundance.**
Cammarota*, D., Zeraick Monteiro*, N., Menezes*, R., ..., Segura, A. M.
Journal of Mathematical Biology, 87, 82
- 2020 **Integrating Computational Methods to Investigate the Macroecology of Microbiomes.**
Mascarenhas, R., Ruziska, F. ... dos Santos, R. M., ..., Meirelles, P.M.
Frontiers in Genetics, 10, 1344
- 2017 **Evaluation of models capacity to predict size spectra parameters in ecosystems under stress.**
dos Santos, R. M., Hilbers, J. P., Hendriks, A. J.
Ecological Indicators, 79, 114-121

* Equal Contribution

Professional Service

- 2025 – present **Assistant Chair** PoLSAN
Assistant Chair of the Physics of Life South American Network.
- 2024 – present **Invited Scientific Reviewer**
Reviewed scientific papers and grants for *Nature Communications*, *Methods in Ecology and Evolution*, *Scientific Reports*, and FAPESP.

Scholarships and Awards

2025 – 2027 (ongoing)	FAPESP Postdoctoral Researcher Independent Postdoctoral Position (24 months)	FAPESP
2023 – 2024	CAPES Internationalization Scholarship Scholarship Grant funded by CAPES (12 months)	CAPES
2021 – 2025	CNPq Ph.D. Scholarship Grant Ph.D. Scholarship Grant funded by CNPq (48 months)	CNPq
2017 – 2019	M.Sc. Scholarship Grant funded by CAPES (24 months)	CAPES
2017	Honorable Mention at the UFBA 2017 Congress Honorable Mention for best student research project: "Dynamical models of vectorial-borne diseases"	UFBA
2016 – 2017	B.Sc. Research Scholarship Grant funded by FAPESB (12 months)	FAPESB
2014 – 2015	Exchange Student – Radboud University, Nijmegen Science Without Borders Exchange Program Grant (12 months)	CAPES
2013 – 2014	B.Sc. Research Scholarship Grant funded by CNPq (12 months)	FAPESB
2012 – 2013	"Young Talents for Science" B.Sc. Research Scholarship Grant funded by CNPq (12 months)	CNPq
2010	Silver Medal at the Brazilian Physics Olympiad Awarded by the State Coordination of the Brazilian Physics Olympics	SBF - BA

Invited Talks and Teaching Experience

2025	Minicourse on Fundamentals of Biological Physics Stochastic Individual Based Models in Theoretical Ecology	(Lecturer) ICTP-SAIJR, Sao Paulo
2025	CASUS Institute Seminar Upscaling realistic animal movement to understand population dynamics	Görlitz, Germany
2024	McManus' Lab Group Meetings Using IBMs to understand space usage and patterns in competitive populations.	Hawaii, USA
2024	Applied Stochastic Processes for Encounter Problems at University of Maryland Upscaling range residency: The range-resident logistic model.	Maryland, USA
2023	Modeling Coffee at Swansea University The scaling of individual home ranges and spatial structure and their effects on the carrying capacity of populations.	Swansea, UK
2023	Eco-Encontros at University of São Paulo Não tão bem misturado: O papel das Menezes, R. 2024. vitais na capacidade de suporte	São Paulo, Brazil
2022	ICTP Huddle - Eco-evolutionary Dynamics of Microbial Communities Across Scales Not so well mixed: the role of range residency in the dynamics of ecological communities	Trieste, Italy

Selected Contributed Talks & Posters

2023	Spatial Ecology Workshop: From animal movement processes to spatial distributions, University of Sheffield, UK Adding realistic movement behavior to spatial models of population growth	Workshop
2022	Applications of Nonlinear Systems to Socio-Economic Complexity What Is and How do Interactions Determine Resilience?	School
2022	Brazilian Physics Society Autumn Meeting Promoting Diversity in Graduate Programs Through Quotas - A Mathematical Model	Conference
2021	Brazilian Physics Society Autumn Meeting Exploring Ecological Interactions Using the Generalized Lotka-Volterra Model	Conference
2021	SMB 2021 Annual Meeting, Society for Mathematical Biology Feasibility and Resilience in Randomly Assembled Communities	Conference

Tutoring & Event Organization

2025	Minicourse on Fundamentals of Biological Physics International scientific event with the participation of 40 students at ICTP-SAIJR.	Organizer
2023, 2025	Training Program in Quantitative Ecology Organizers: ICTP-SAIJR (2023), Serrapilheira Institute (2023, 2025).	Tutor
2022	II Encontro da Pós Graduação IB-USP Scientific event with the participation of approx. 90 people, at University of São Paulo	Organizer
2022	Escola de Biomatemática da Bahia Organizers: Flora Souza Bacelar, Roberto Kraenkel, Paulo Inácio Prado, Gilson Correia de Carvalho	Tutor
2019 - 2020	VIII - IX Southern Summer School of Mathematical Biology Organizers: Marcus Aguiar, Marcel Clerc, Roberto Kraenkel, Paulo Inácio Prado	Tutor